Vivo first Android 11 phone
Vivo has beaten Google to the punch with the launch of the new Vivo V20. http://dgit.in/oct20-85

Space toilet
NASA is preparing to send its $23 million toilet to space. It'll be the first new loo since the 1990s. http://dgit.in/oct20-86

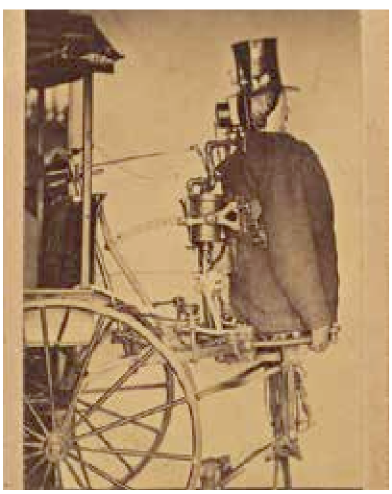
steam powered vehicles were made during this time, many of which continue to be maintained and serviced by enthusiasts. When the impact of emissions and greenhouse gases were better understood around the 1960s, there was a renewed interest in steam powered cars. Hobbyists retrofitted steam powered engines into existing vehicles, and auto manufacturers came up with the occasional prototype or concept design, based on steam power. One British pensioner, Frank Rothwel took an old Land Rover, and spent 400 hours adding a slower, and noisier steam engine. The steam is generated by burning coal, and it requires about 45kg of coal every hour. Apart from the enthusiasts and the odd experiments by manufacturers, one driving force for steam powered vehicles was the land speed record for a steam powered vehicle, which was a surprisingly fast 225.055 km/h set by a Stanley Steamer in 1906. The record itself set the record of being the longest standing land speed record, with the timing being undefeated for over a century. It was only in 2009, that a steam powered car known as Inspiration, managed to break the steam powered land speed record, by reaching a top speed of 238.679 km/h. The funny thing is that the vehicle burns LPG to produce the heat needed to generate the steam. Inspiration is 7.6 m long, 1.7 m wide, and 1.7 m tall. The burners generate enough heat to make 12 cups of tea every second, and the steam is routed through 3 kilometers of tubing.

Credit: steamcar.co.uk

Inspiration, the vehicle that holds the steam powered land speed record

ROBOTS

Between 400 to 350 BCE, there is evidence that the Greek mathematician Archytas constructed a robotic pigeon out of wood, which used a jet of steam for flying. It could apparently fly 200 meters before it ran out of steam. In more modern times, there were all kinds of ridiculous inventions built using steam power, from the very early days of steam engines. In 1868, a pair of American inventors Zadoc P Dederick and Isaac Grass created a steam powered android to pull a cart. The robot moved the cart forward through a bipedal motion, similar to the way a man walks. The unlikely consideration for making the robot roughly human in shape, was so that the horses, which were far more common at that time, would not be spooked by what was essentially a seven foot tall mechanical scarecrow bellowing steam. This man had a moustache, gloves, a top hat and a jacket, but rather thin and spindly legs, and no pants. Of course, Mr Steam Man had a steam whistle in his mouth. Dederick had plans to make a steam powered mechanical horse as well. Unfortunately, there were not too many takers for the $300 steam men. I-Wei Huang, a video game concept artist who goes under the pseudonym CrabFu has made a number of steam powered hobbyist robots. These are mostly remote controlled steam powered vehicles of various types, including tanks, boats, cars and mechanical insects. There is even a protocol droid called R2S2 (the S stands for steam, obviously). You can check out more at crabfu.com.



Dederick's Steam Man

SPACECRAFT

NASA's engineering team has come up with a concept spacecraft that could one day explore the ice worlds in the Solar System. The design is particularly suitable for the low gravity environments on the Saturnian moon of Enceladus, and the Jovian moon of Europa, both of which have subsurface oceans, and the potential for harbouring life. There are two advantages of using the steam powered probe that can hop away from the landing site. The first is that there is plenty of water available for converting into steam, considering that the moons are covered by ice shells. The probe mines the ice it needs to fly at the location. The second is that by not using rocket fuels, the environment through which the probe navigates is preserved for study. The concept is called Steam Propelled Autonomous Retrieval Robot for Ocean Worlds or SPARROW.

No matter what surprises can be found in these ice worlds, including tortured terrain, soft snow, crevasses and penitentes, SPARROW's design is capable of navigating through the environment. NASA has also funded a project by a startup known as Honeybee Robotics and researchers from the University of Central Florida. This autonomous steam powered spacecraft can mine water from objects in the solar system, and then use that as a propellant. The technology has been demonstrated with a prototype and simulated asteroids. Water is abundant in the solar system, with even Mercury preserving some in the polar ice caps. Such a spacecraft could potentially jump from the Earth to the Moon, to Europa, make a pitstop on an asteroid and go all the way to Pluto before refuelling and heading inwards. In fact, a spacecraft that runs on water could keep travelling across the Solar System practically forever.